

GLUCONEOGENESIS AND CONTROL OF BLOOD GLUCOSE

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Objectives:

- At the end of this lecture, it is expected to;
 - Know the Biomedical Importance of Gluconeogenesis
 - Describe the pathway of Gluconeogenesis in details
 - Understand the differences and similarities between gluconeogenesis and glycolysis
 - Understand the regulations of Gluconeogenesis -reciprocal control of glycolysis and gluconeogenesis
 - Calculate the energetics of gluconeogenesis
 - Describe Cori cycle and understand the relation between gluconeogenesis and glycolytic pathway
 - Describe the mechanism of Blood glucose homeostasis and relate the gluconeogenesis and control of Blood glucose
 - Describe the metabolism of alcohol and understand its effect in gluconeogenesis and fuel metabolism

Biomedical Importance:

- To meet the requirements of glucose in the body **when carbohydrates are not available in sufficient amounts from the diet.**
- Even in conditions, where fat is utilized for energy still certain basal level of glucose is required to meet the need of special uses, e.g.
 - **Source of energy for brain (nerve tissues) and erythrocytes,**
 - Required for **maintaining level of intermediates of TCA cycle,**
 - **Source of glyceride-glycerol-P** required for adipose tissue,
 - As **precursor of milk sugar** (lactose) for lactating mammary gland,
 - **As only fuel for skeletal muscles** in anaerobic conditions.
- To clear the products of metabolism of other tissues, e.g.
 - **Lactic acid** produced by muscles and erythrocytes,
 - **Glycerol** which is continuously produced by adipose tissue by lipolysis of TG (triacyl glycerol).

Glucose is an indispensable metabolite

- The brain requires at least ~50% of its calories in the form of glucose
- Red blood cells exclusively subsist on glucose
- Glucose is a precursor of other sugars needed in the biosynthesis of;
 - Nucleotides,
 - Glycoproteins, and
 - Glycolipids
- Glucose is needed to replenish NADPH
 - For supply of reducing power for biosynthesis and detoxification

Biomedical Importance:

- **Blood glucose levels are maintained within a very narrow range;**
 - Even though the nature of a person's diet may vary widely, and
 - The normal person eats periodically during the day and fasts between meals and at night.
 - Even under circumstances when a person does not eat for extended periods of time, blood glucose levels decrease only slowly.

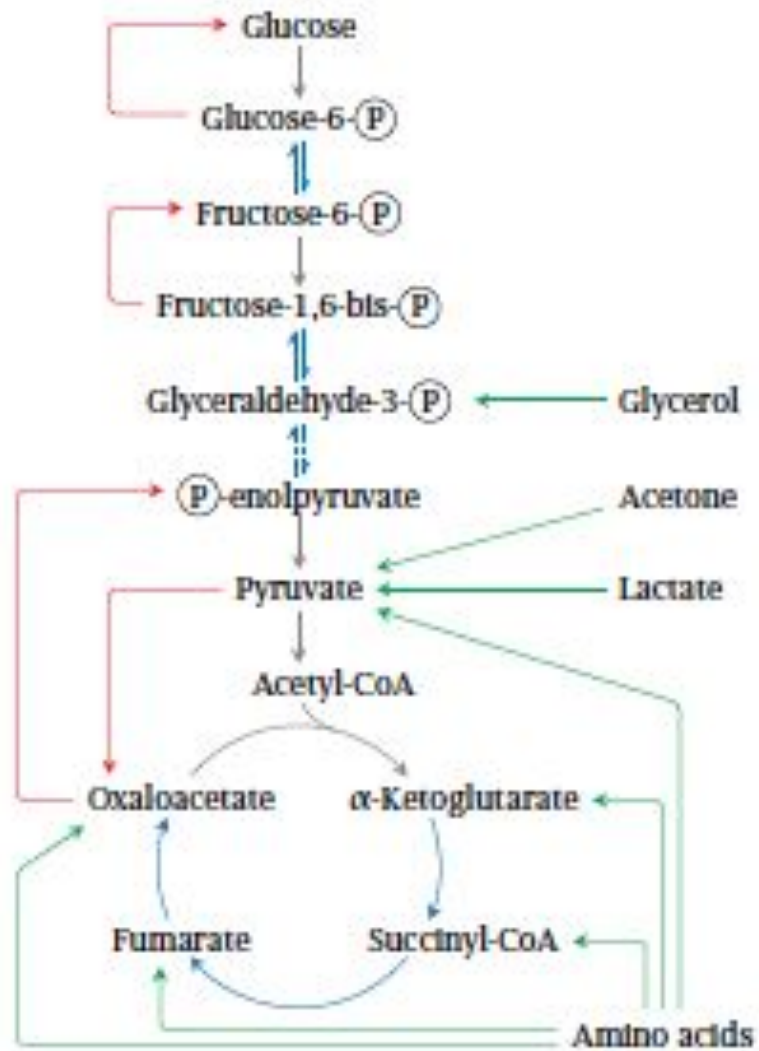
GLUCONEOGENESIS

- **Synthesis of glucose from compounds that are not carbohydrates.**
- Occurs mainly in the **liver** and to a small degree in the kidney.
- The major precursors are:
 - Pyruvate,
 - lactate,
 - Some amino acids skeleton
 - Glycerol
 - TCA cycle intermediates
- Even-chain fatty acids do not produce acetyl-CoA
- Involves glycolysis, the citric acid cycle, & some special reactions.
- Not simple reversal of glycolysis; Involves several enzymatic steps that do not occur in glycolysis;
 - Thermodynamic Barriers Prevent a Simple Reversal of Glycolysis.
- The **synthesis of 1 mole of glucose from 2 moles of pyruvate** requires about 6 ATPs.

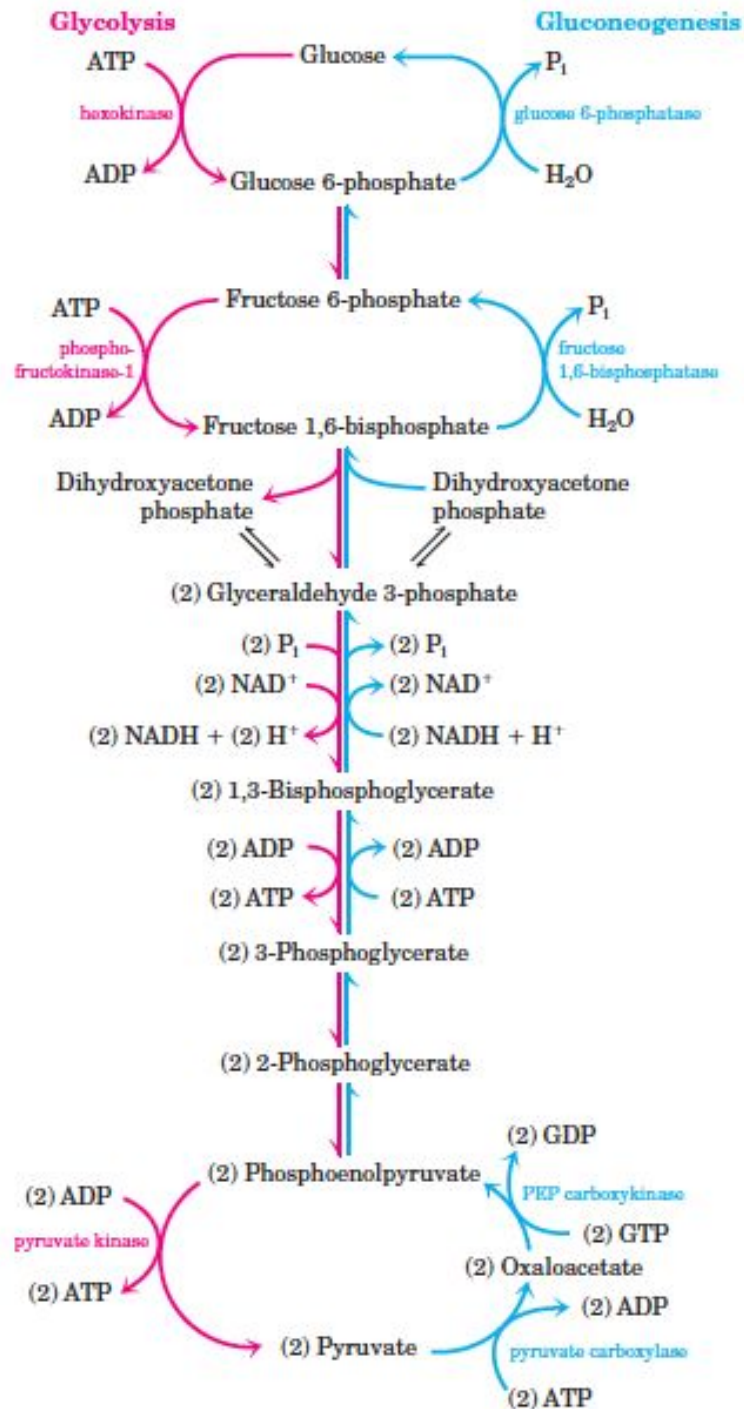
List of amino acids which are glucogenic:

- **Pyruvate forming amino acids are:** Glycine, alanine, serine, cysteine, cystine, and threonine.
- **Oxaloacetate forming amino acids:** Aspartic acid.
- **α -Oxoglutarate forming amino acids are:** Glutamate, Glutamine, Proline, Arginine, Histidine, Lysine.

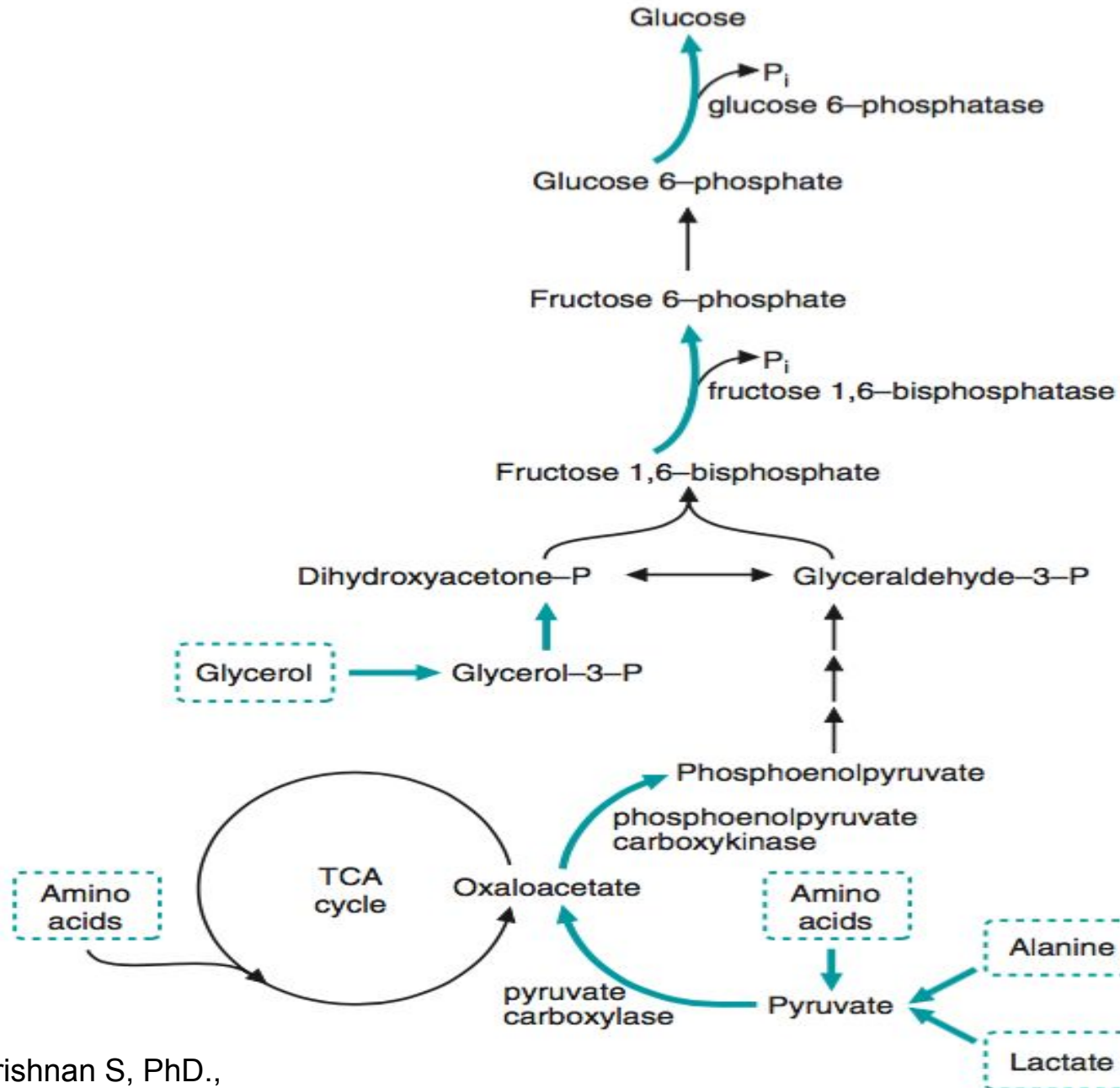
OVERVIEW

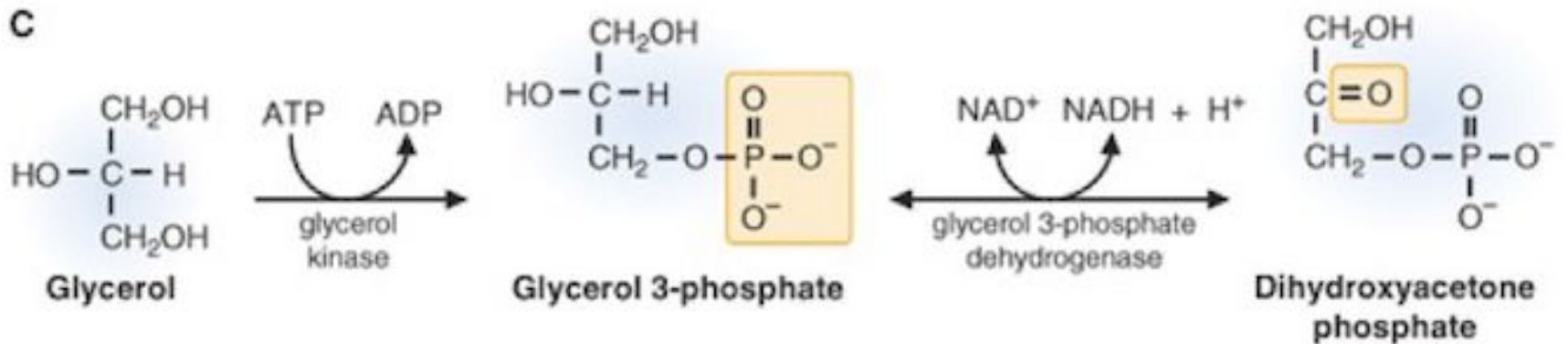
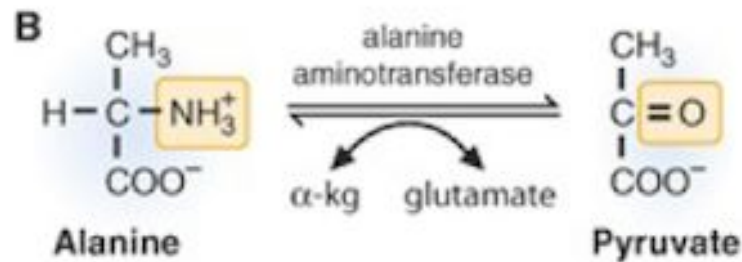
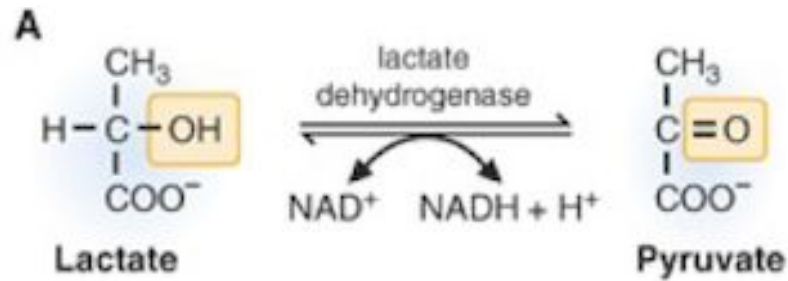


Opposing pathways of glycolysis and gluconeogenesis in rat liver.



Key reactions of gluconeogenesis from the precursors (alanine, lactate, and glycerol)



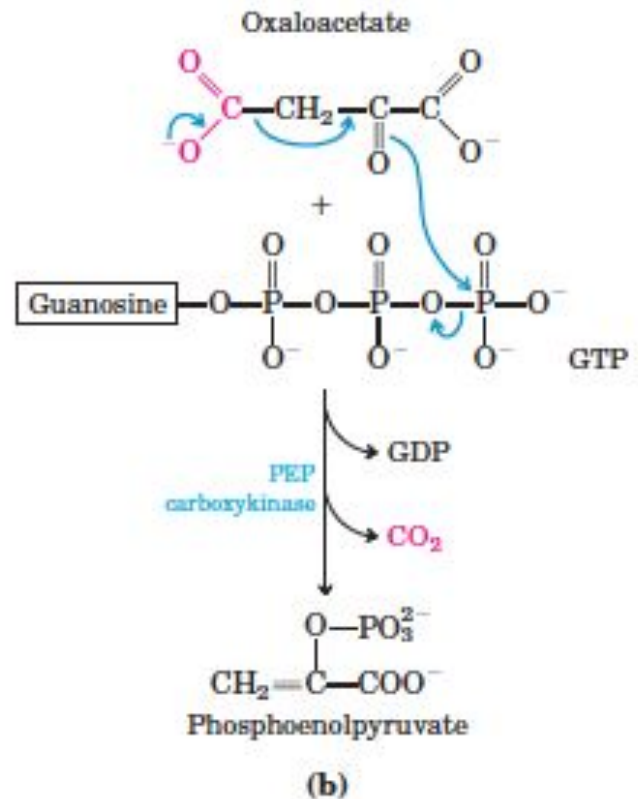
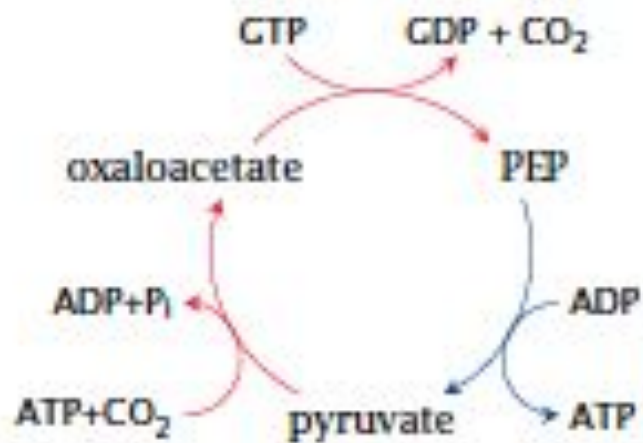
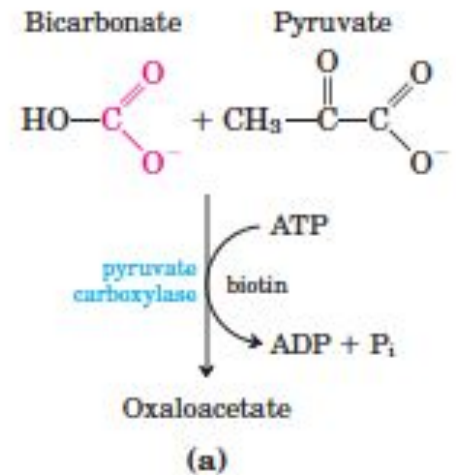


Bypass number 1:

Pyruvate to phosphoenolpyruvate.

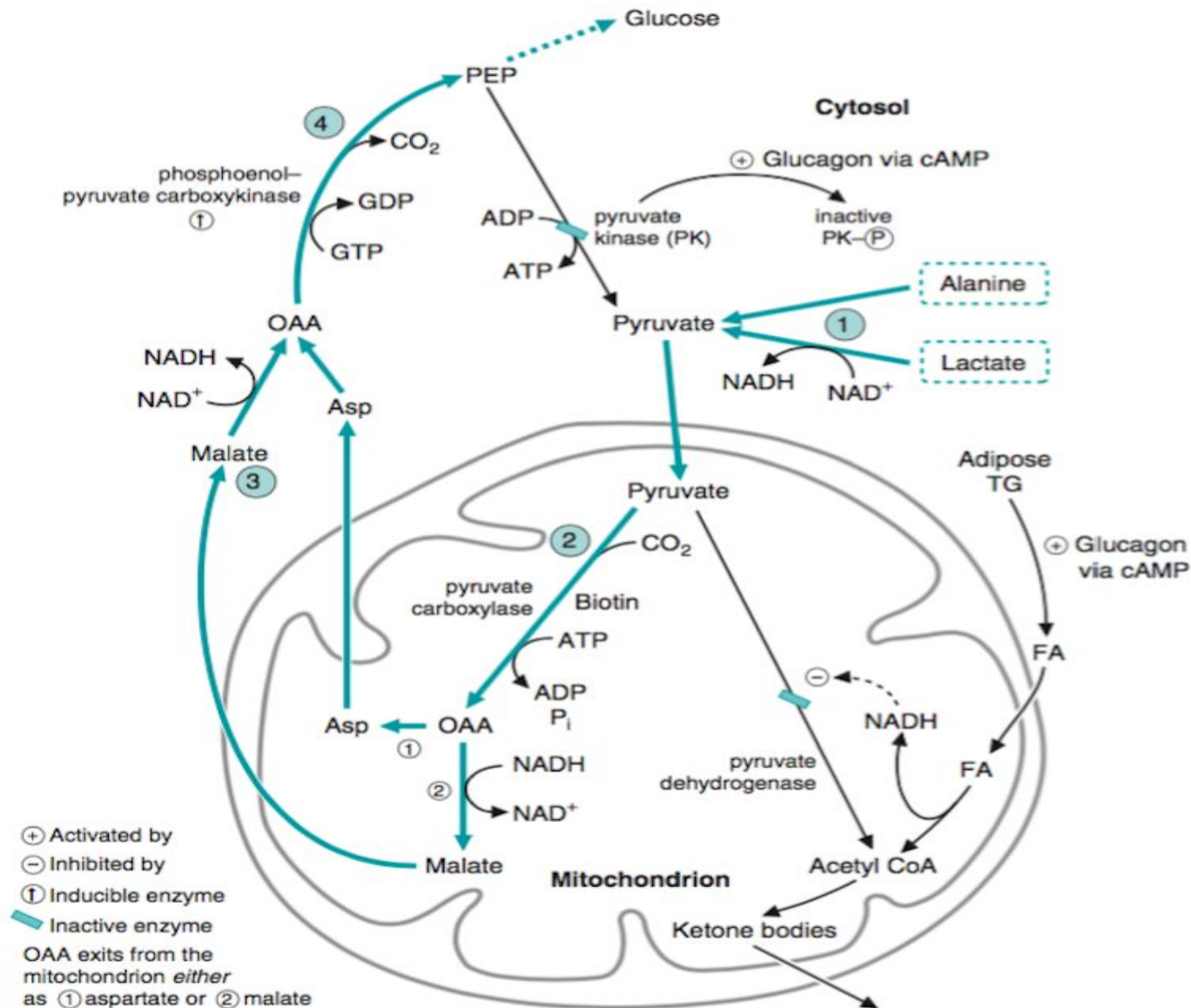
This bypasses pyruvate kinase.

☞ Complex scheme.



- (a) In mitochondria, pyruvate is converted to oxaloacetate in a biotin requiring reaction catalyzed by pyruvate carboxylase.
- (b) In the cytosol, oxaloacetate is converted to phosphoenolpyruvate by PEP carboxykinase.

Conversion of pyruvate to phosphoenolpyruvate(PEP)



CLINICAL CORRELATES

PEPCK deficiency is a rare but severe metabolic defect. Absence of the cytosolic form of the enzyme results in severe **cerebral atrophy**, optic atrophy, **fatty infiltration** of the liver and kidney, and **intractable hypoglycemia**.

Dicarboxylic acid Shuttle

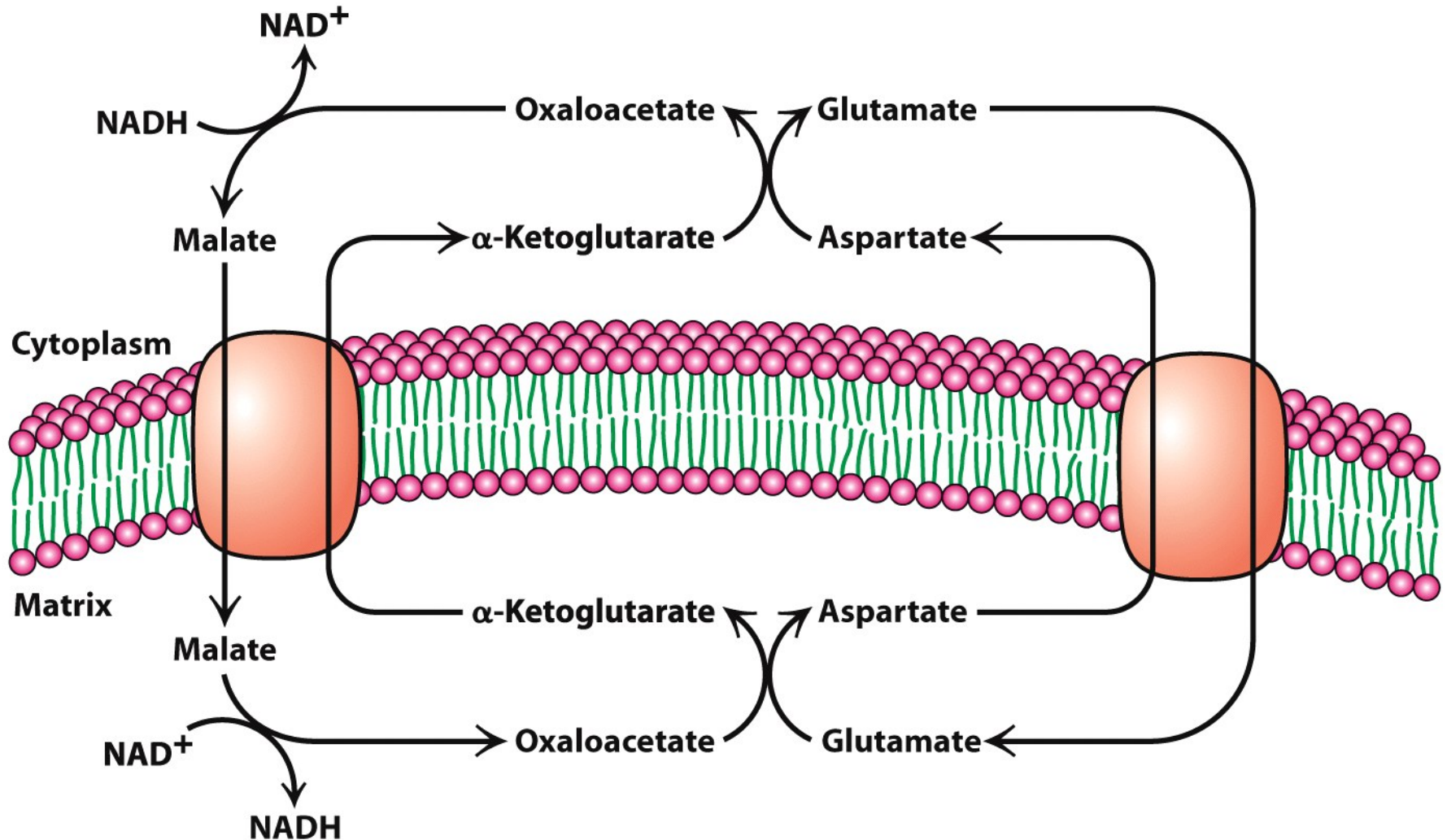


Figure 18.35

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Bypass number - 2: Fructose-1,6-*bis*phosphate to fructose-6-phosphate

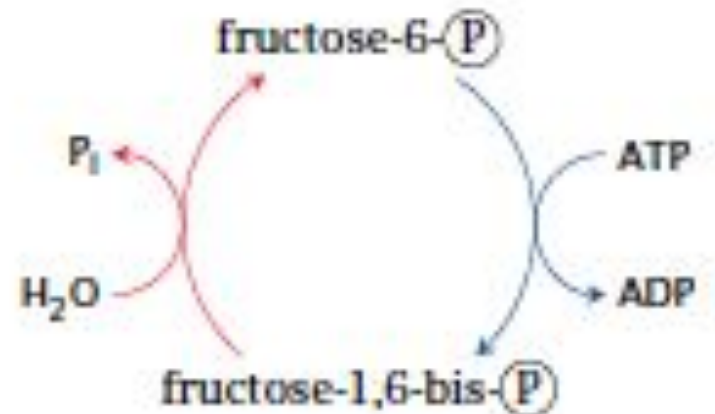
Enzyme = **fructose-1,6-*bis*phosphatase**

Reaction:



- bypasses phosphofructokinase
- a simple hydrolysis.
- highly exergonic, irreversible
- enzyme is highly regulated

F6P isomerizes to glucose-6-phosphate via phosphoglucoisomerase



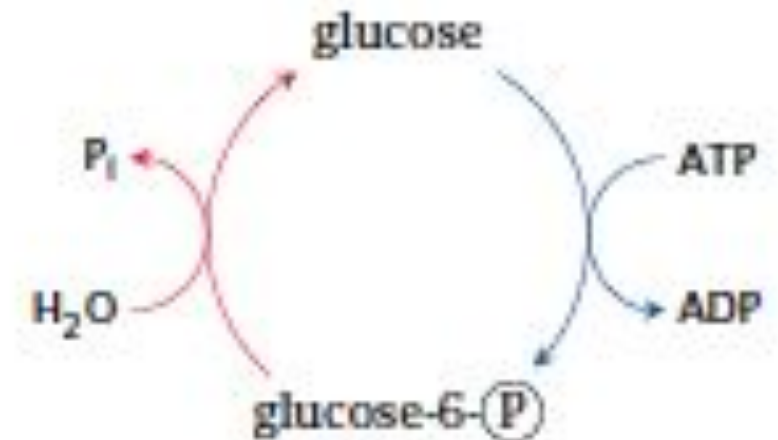
Bypass number – 3: Glucose-6-phosphate to glucose.

Enzyme = **glucose-6-phosphatase**

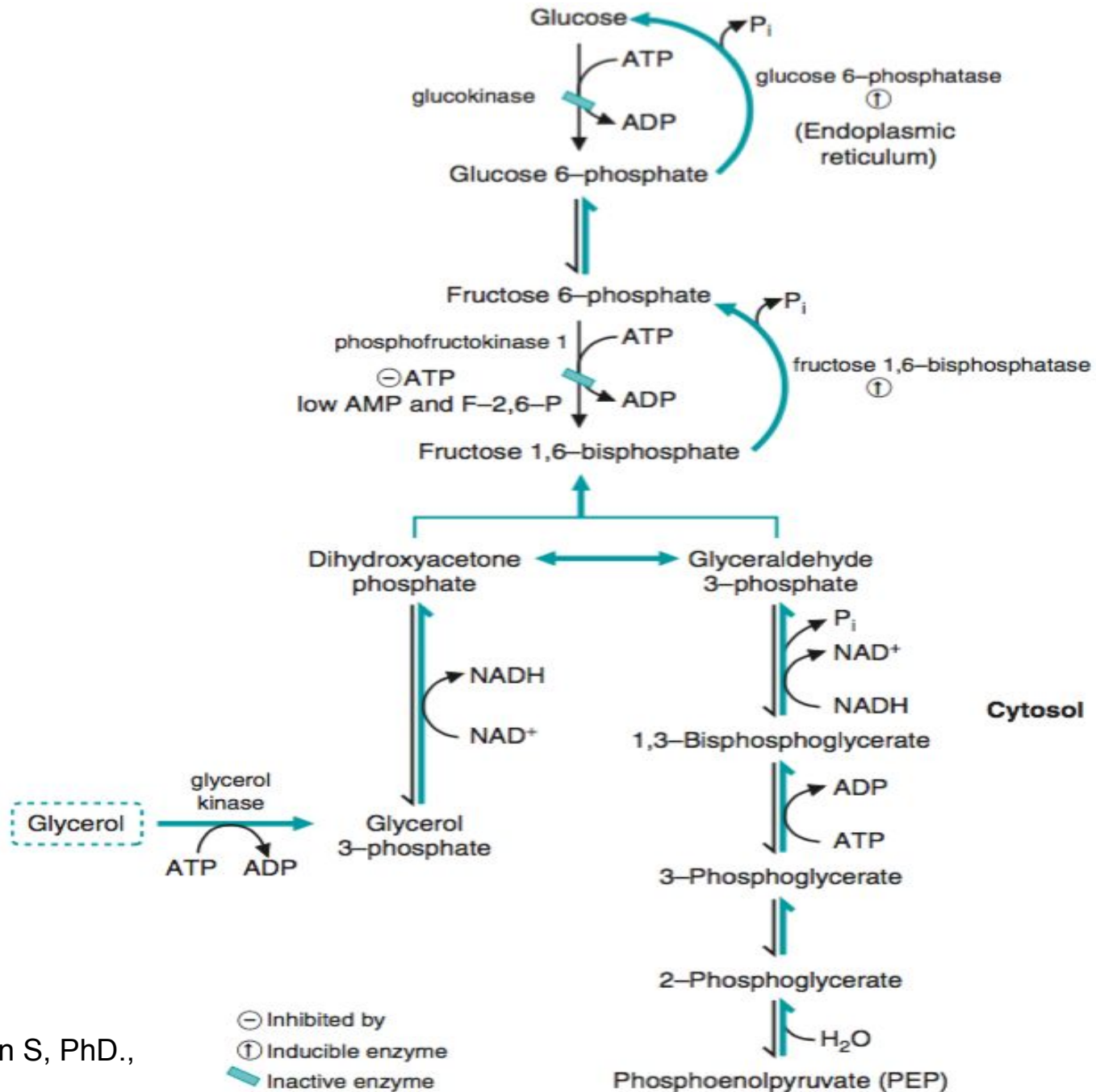
Reaction:



- bypasses hexokinase
- highly exergonic, irreversible
- not present in muscle



Conversion of phosphoenolpyruvate and glycerol to glucose



CLINICAL CORRELATES

Deficiency of **glucose 6-phosphatase, von Gierke disease**, is a type I glycogen storage disease (GSD). Failure to convert glucose 6-phosphate to glucose results in intracellular accumulation of glucose 6-phosphate and **severe hypoglycemia** that can cause **lethargy, seizures, and brain damage**. Patients often have **hepatomegaly**, increased bleeding (due to platelet dysfunction), and **growth retardation**. **Frequent meals** and nighttime nasogastric feedings help control the disease.

CLINICAL CORRELATES

Deficiency of **F-1,6-bisphosphatase** commonly presents as **neonatal hypoglycemia**, along with **acidosis**, irritability, tachycardia, dyspnea, **hypotonia**, and **moderate hepatomegaly**. This deficiency is typically only of the liver enzyme, and the **muscular F-1,6-BP activity is normal**.

CLINICAL CORRELATES

Genetic **deficiency of lactate dehydrogenase** has been described. These patients present with **muscle cramping and myoglobinuria after intense exercise**.

ENERGETICS

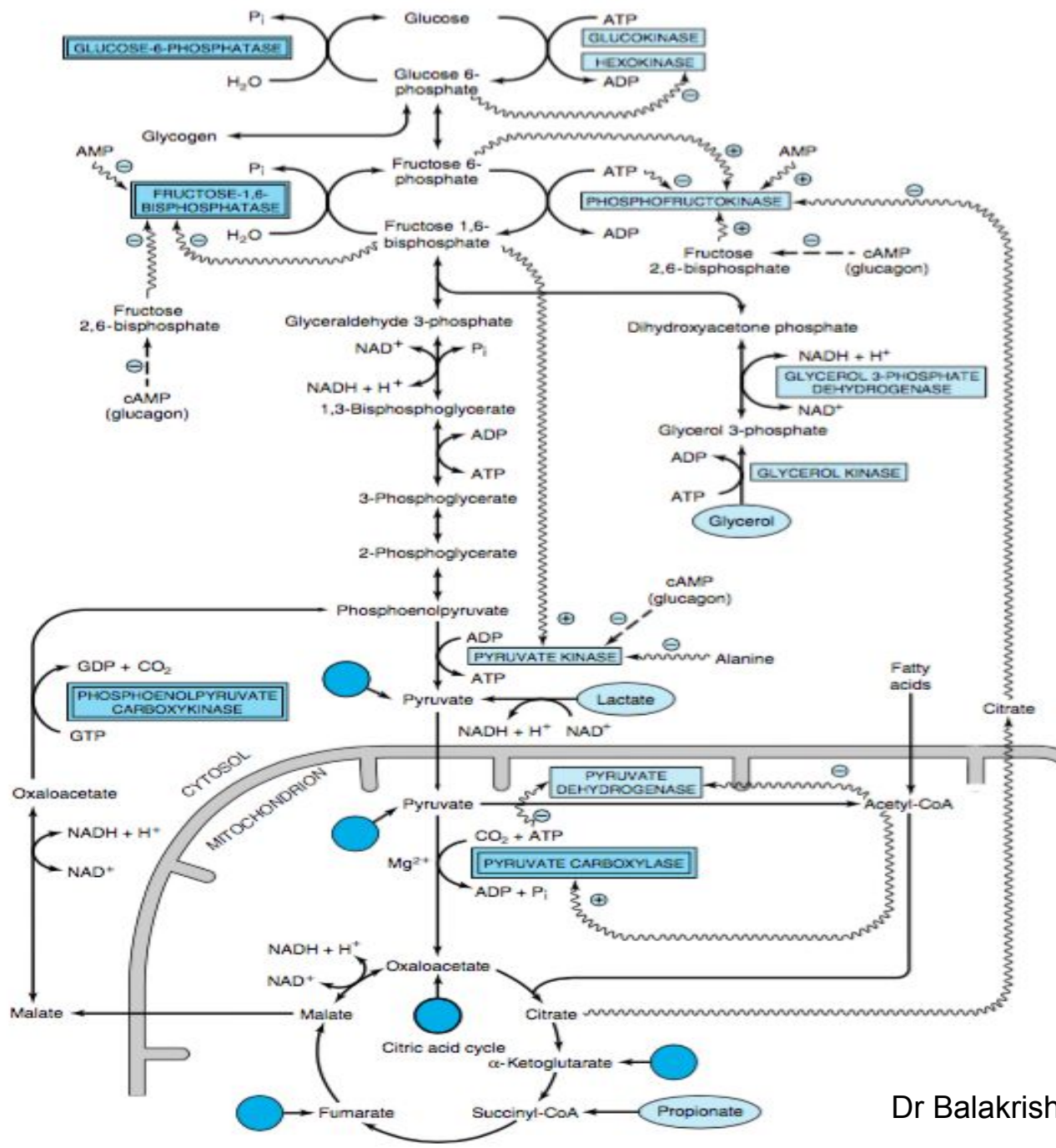
Synthesis of 1 mole of glucose from 2 moles of pyruvate:

- ***Total Energy Cost*** = 6 high energy bonds used.
- Four more than produced in glycolysis.
- These four are needed to convert pyruvate to PEP.

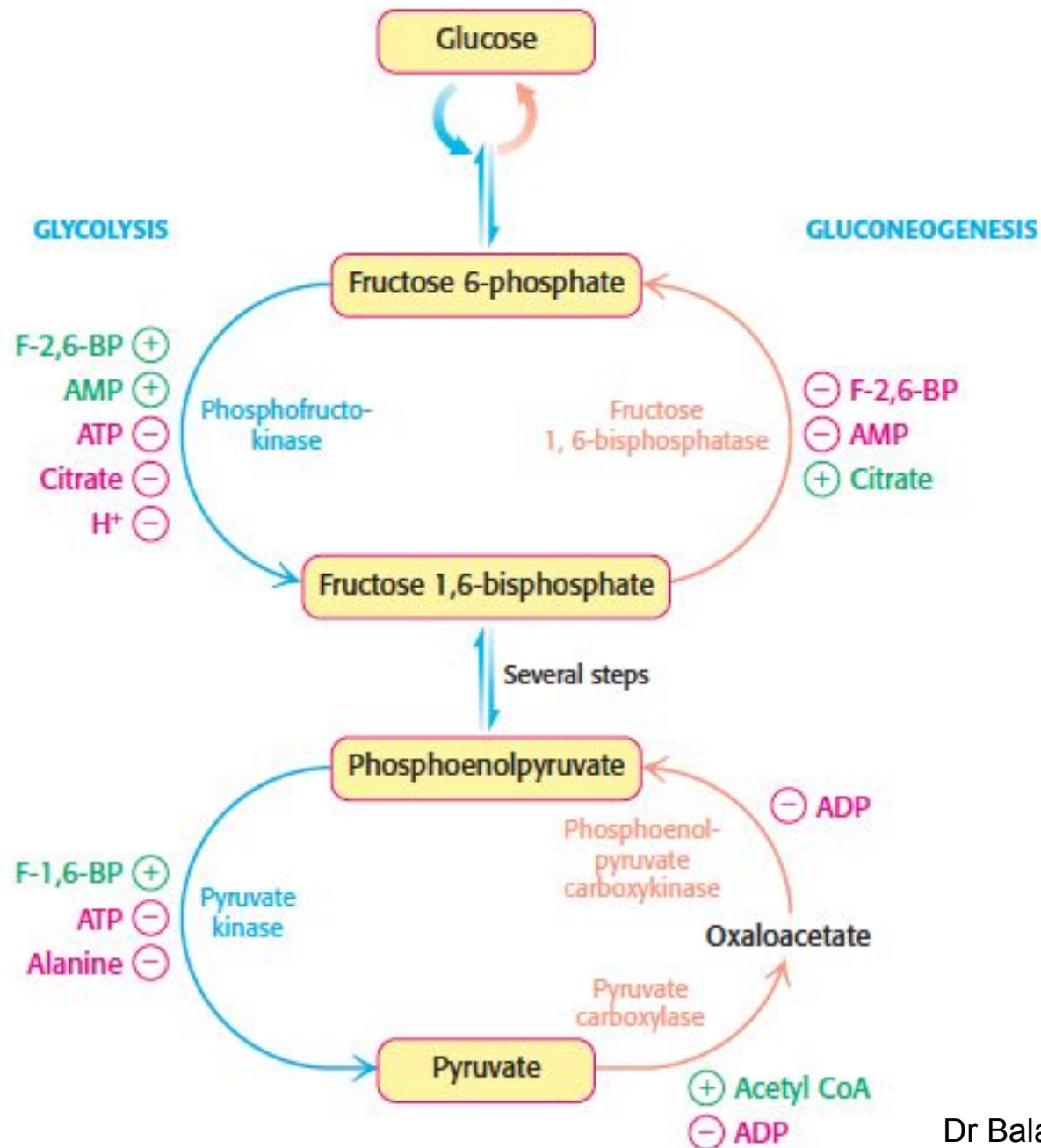
CONTROL OF BLOOD GLUCOSE Vs GLUCONEOGENESIS:

- Gluconeogenesis serves as an alternative source of glucose when supplies are low and is largely controlled by diet.
- High carbohydrate in meal reduce gluconeogenesis and starvation increases.
- Gluconeogenesis and glycolysis are controlled in reciprocal fashion.

Reciprocal regulation of gluconeogenesis and glycolysis



The reciprocal regulation of gluconeogenesis and glycolysis in the liver.



REGULATION OF GLUCONEOGENESIS:

Key enzymes:

1. PYRUVATE CARBOXYLASE.

- *Activated* by acetyl-CoA (Pyruvate kinase - *inhibited* by acetyl CoA).
- High levels of acetyl-CoA signals that enough carbon substrate available for citric acid cycle.
- Pyruvate kinase is also inhibited by ATP.
- Inhibited by Avidin;
 - Biotin binding protein found in raw eggs.

Case file: Pyruvate Carboxylase Deficiency

- A 6-month-old infant presents with developmental delay, failure to thrive, and metabolic acidosis. Laboratory tests reveal elevated lactate, pyruvate levels, ketosis/ketonuria, and hyperammonemia.
- Biochemical changes:
 - Impaired gluconeogenesis
 - Defects in the Krebs cycle.
 - The intermediates of the tricarboxylic acid (TCA) cycle are decreased, leading to metabolic acidosis, elevated lactate, and pyruvate levels.
 - Hyperammonemia is also present due to impaired urea cycle function.
- The patient's management:
 - Supportive treatment, including dietary management, supplementation with carnitine and biotin
 - Management of metabolic acidosis.
 - Regular monitoring of symptoms and complications is essential to optimize the patient's quality of life and prevent further metabolic derangements.

2. FRUCTOSE 1,6 *B*ISPHOSPHATASE

- *Strongly inhibited* by AMP
- *Strongly inhibited* by fructose- 2,6-bisphosphate
- **Fructose-2,6-bisphosphate:**
 - ***Most important regulator*** of glycolysis and gluconeogenesis through its *reciprocal effects* on fructose 1,6-bisphosphatase and phospho-fructokinase.
 - F-2,6-BP synthesis is controlled by cAMP
 - F-2,6-BP is formed from F-6-P by PFK-2

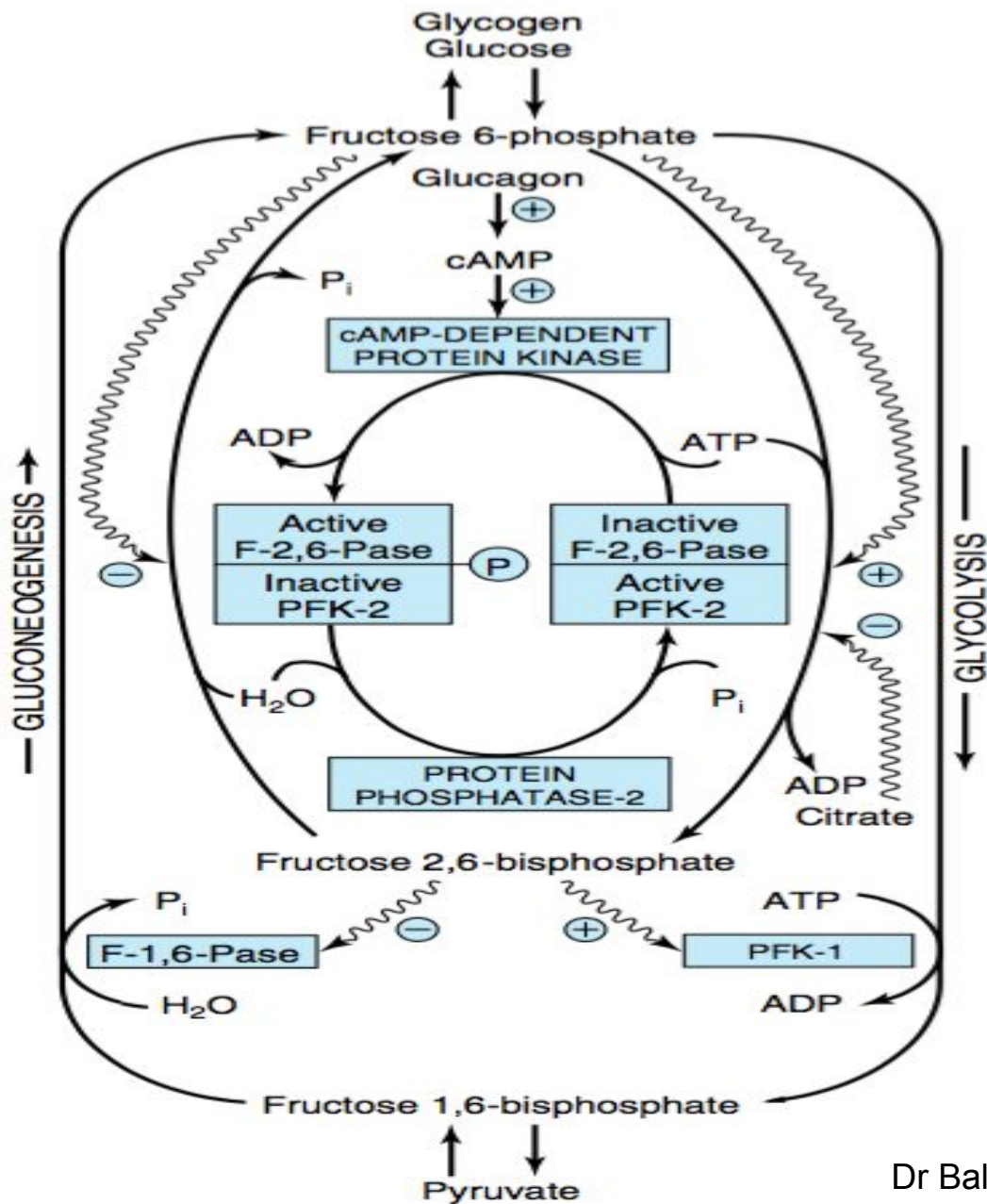
3. **PHOSPHOFRUCTOKINASE-2/FRUCTOSE *B*ISPHOSPHATASE-2** (Bifunctional enzyme)

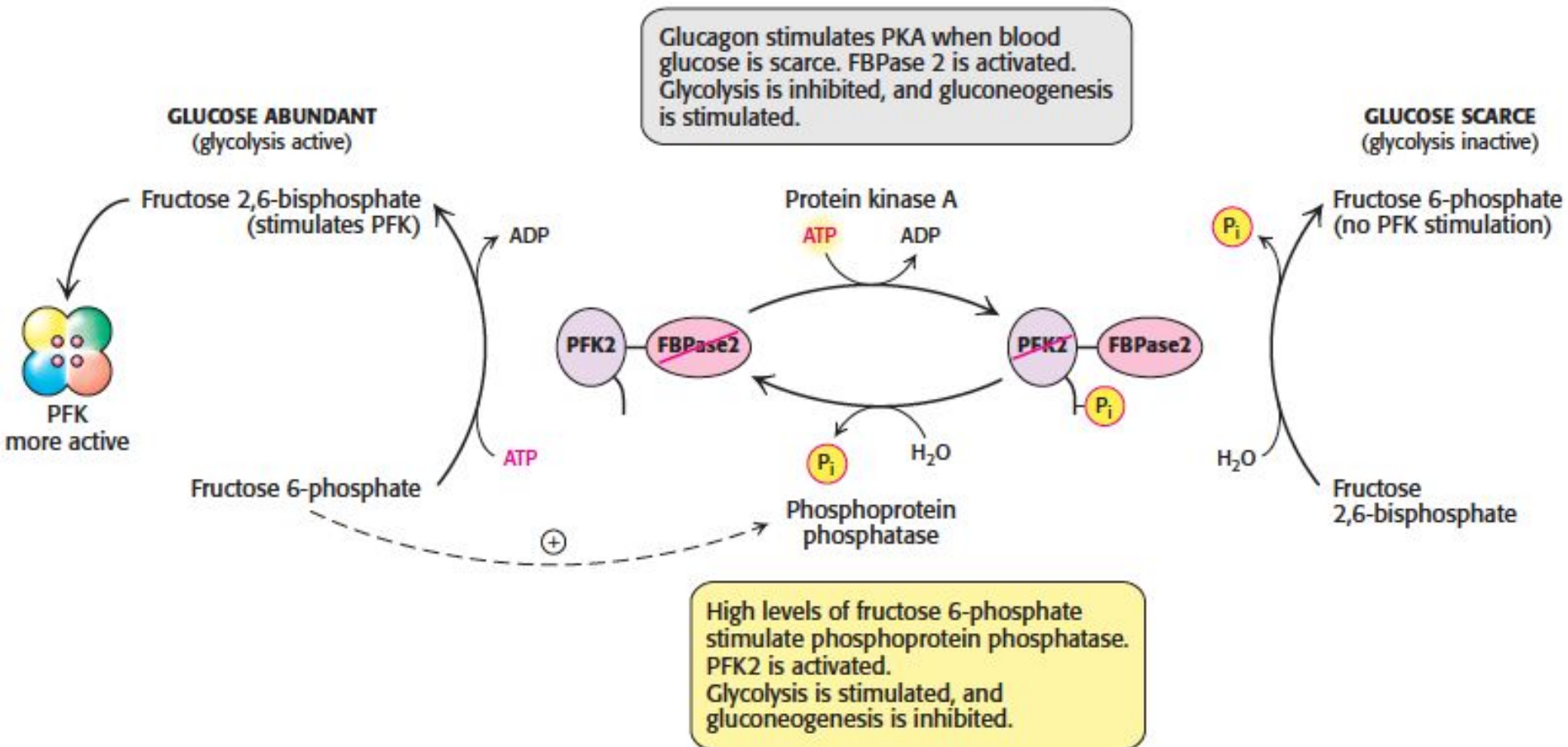
- ① fructose-6-phosphate + ATP $\rightleftharpoons$ fructose-2,6-*bis*phosphate
- ② fructose-2,6-*bis*phosphate $\rightleftharpoons$ fructose-6-phosphate + P_i

Enzyme highly regulated to control levels of F 2,6-P

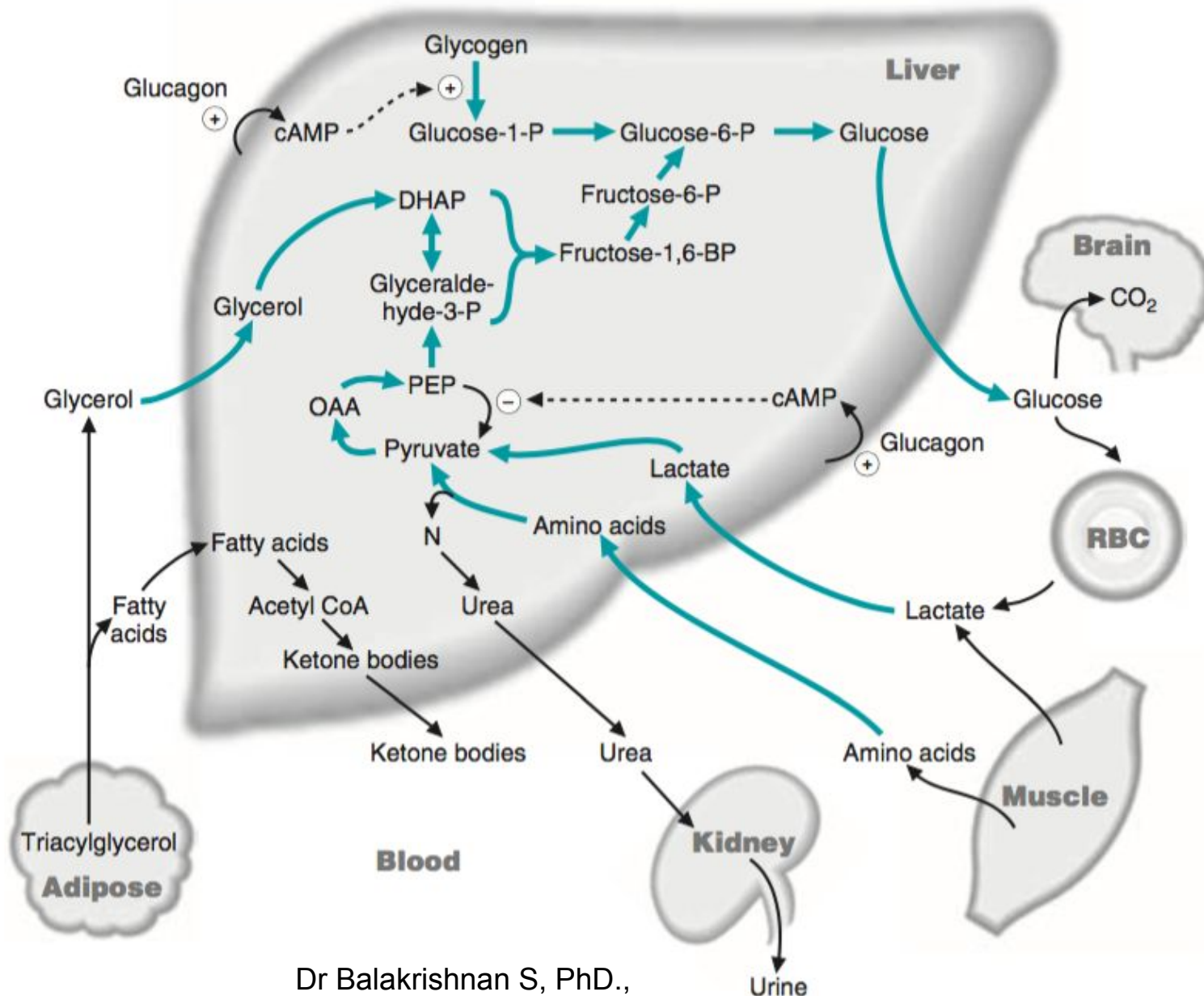
- In the heart, cAMP activates rather than inhibits PFK-2, since gluconeogenesis is not possible.
- In skeletal muscle cell has no cAMP regulation site.
- In the liver, cAMP inhibits PFK-2, activating FBPase-2

Control of glycolysis and gluconeogenesis in the liver

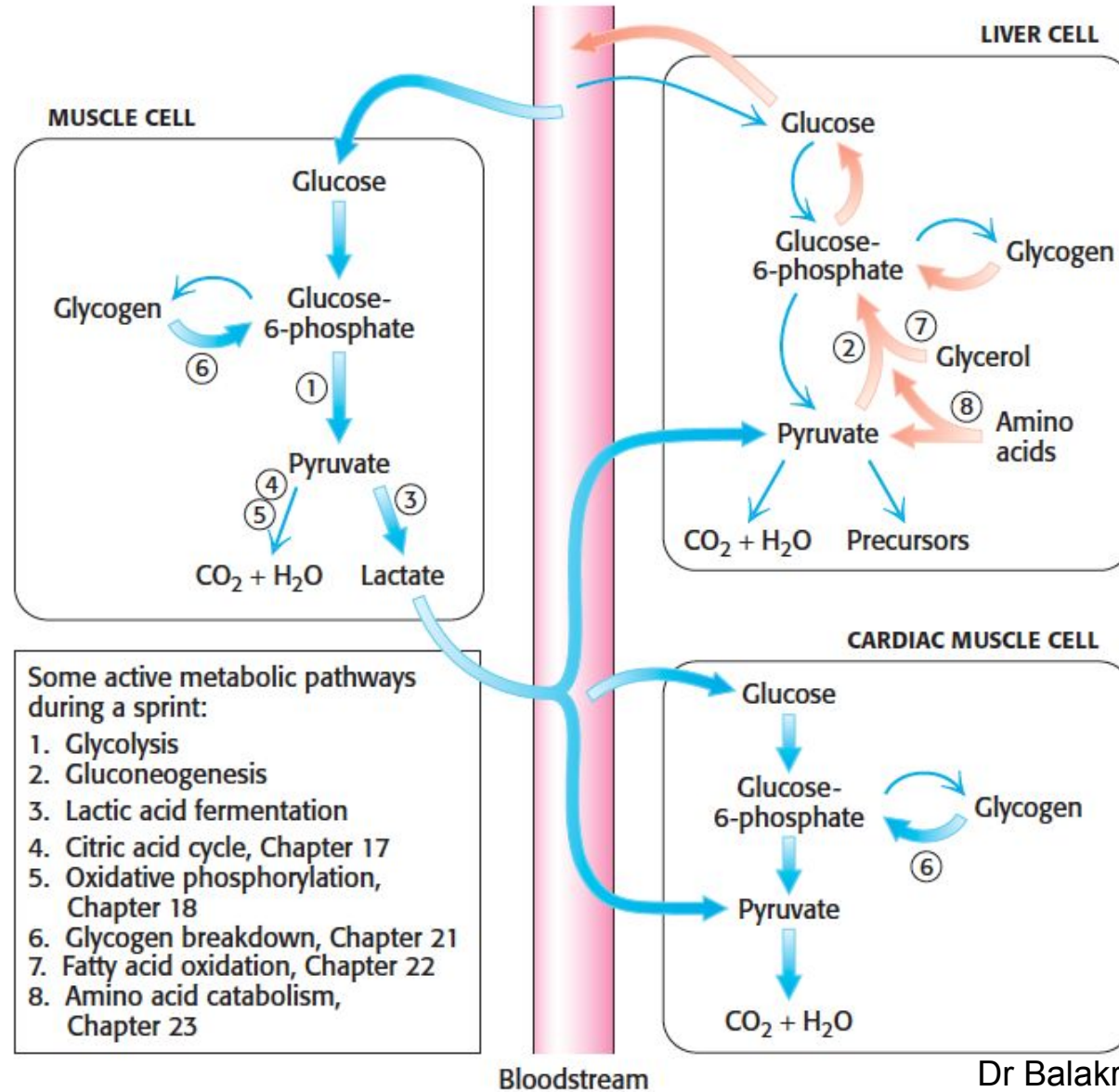




Tissue interrelationships in glucose production during fasting



Cooperation between glycolysis and gluconeogenesis during a sprint or vigorous exercise



Cori cycle

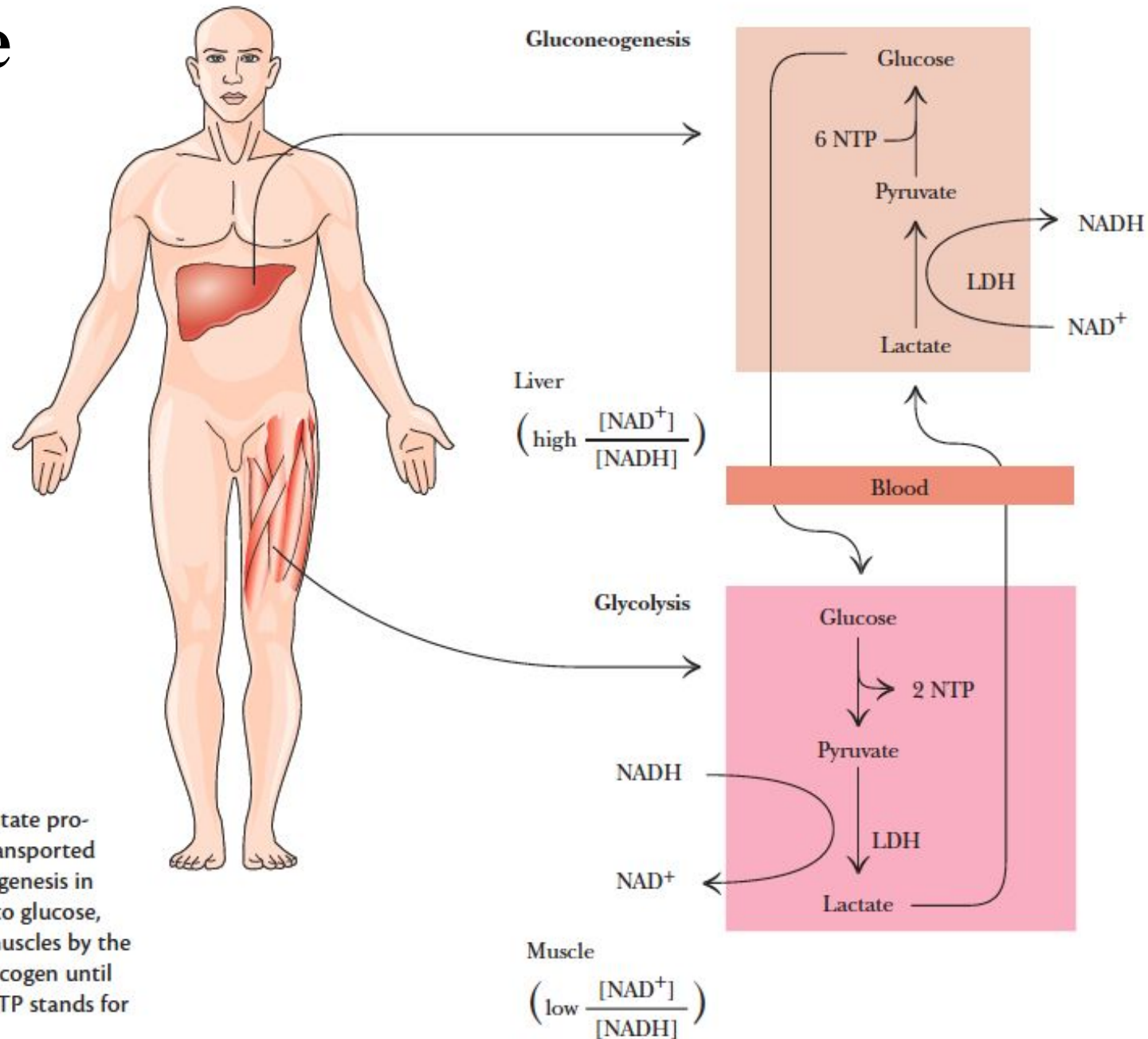
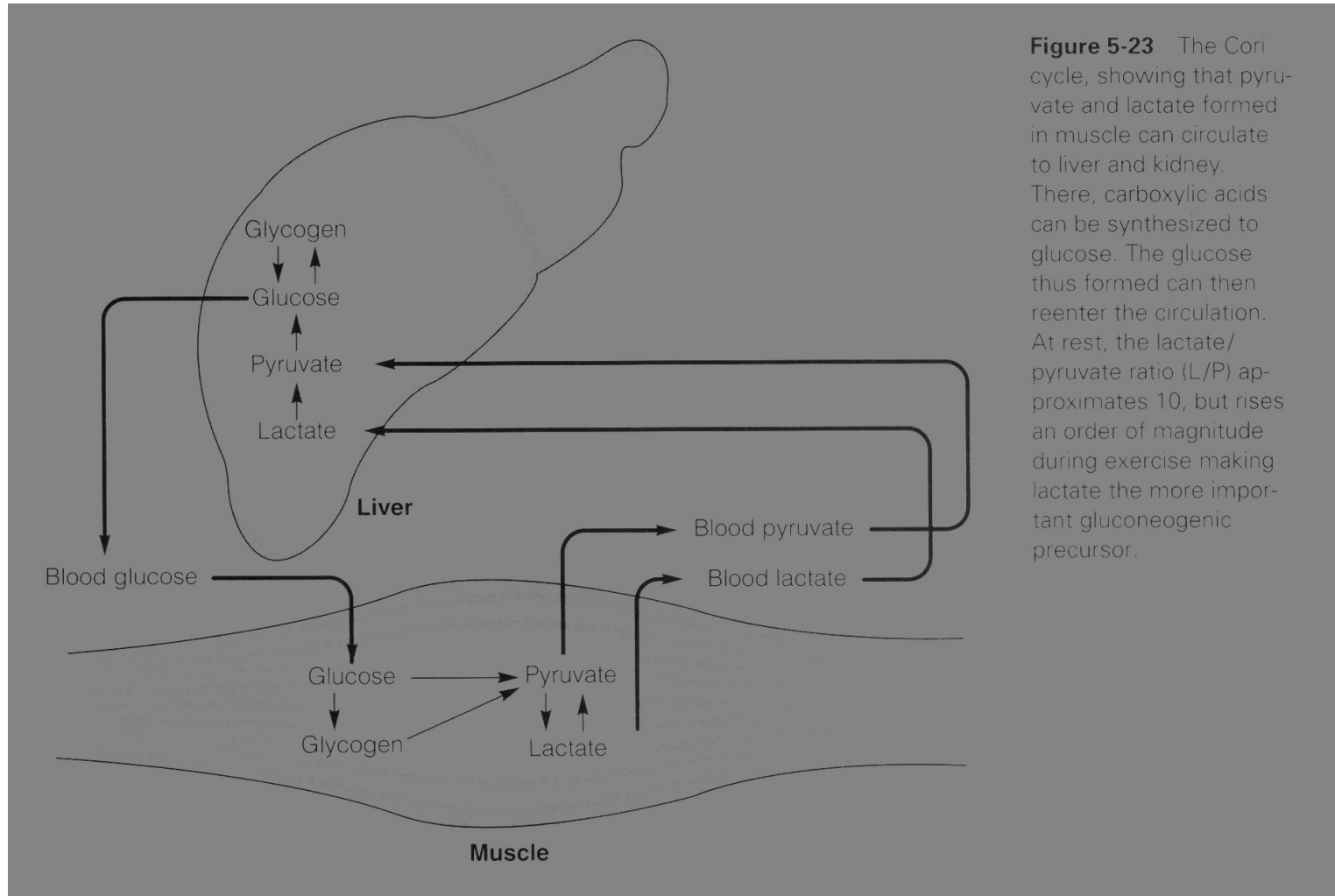


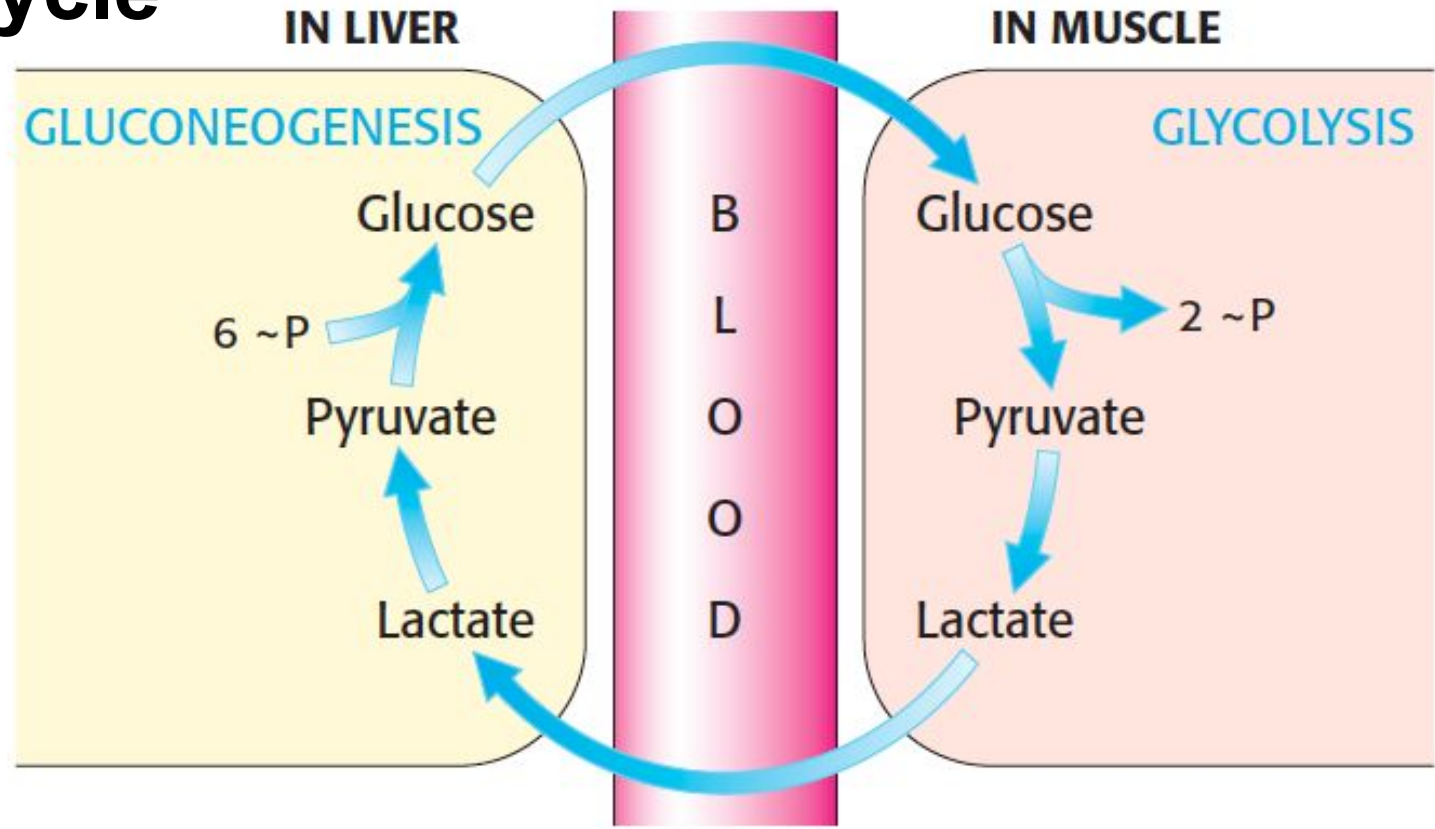
FIGURE 18.17 The Cori cycle. Lactate produced in muscles by glycolysis is transported by the blood to the liver. Gluconeogenesis in the liver converts the lactate back to glucose, which can be carried back to the muscles by the blood. Glucose can be stored as glycogen until it is degraded by glycogenolysis. (NTP stands for nucleoside triphosphate.)

Glycolysis

Cori cycle



The Cori cycle



Substrates for Gluconeogenesis: **Lactate, pyruvate, glycerol, propionny-CoA and certain Amino Acids but never FAT!!!**

The Cori cycle involves the utilization of lactate, produced by glycolysis in non-hepatic tissues, (such as muscle and erythrocytes) as a carbon source for hepatic gluconeogenesis. In this way the liver can convert the anaerobic byproduct of glycolysis, lactate, back into more glucose for reuse by non-hepatic tissues. Note that the gluconeogenic leg of the cycle (on its own) is a net consumer of energy, costing the body 4 moles of ATP more than are produced during glycolysis. Therefore, the cycle cannot be sustained indefinitely.